

#####

STARTING CONDITION B TEST RUN

Total problems: 15

#####

#####

Running problem 1/15

#####

=====
=====

CONDITION B: Classification Only: Topic/domain metadata, no method card guidance

Problem ID: P01

Problem: Add $\frac{1}{2} + \frac{1}{5}$

=====
=====

Classification Signal from Elasticsearch

Score: 0.69088745 | Topic: m_mixed_number | Domain: fractions-decimals | Doc:
fractions-decimals__m_mixed_number-p002

Score: 0.66497064 | Topic: f_fractions_addition | Domain: fractions-decimals | Doc:
fractions-decimals__f_fractions_addition-p002

Score: 0.6509206 | Topic: f_fractions_addition | Domain: fractions-decimals | Doc:
fractions-decimals__f_fractions_addition-p001

$$\begin{aligned}\frac{1}{2} + \frac{1}{5} &= \frac{1 \cdot 5}{2 \cdot 5} + \frac{1 \cdot 2}{5 \cdot 2} \\ &= \frac{5}{10} + \frac{2}{10} \\ &= \frac{5 + 2}{10} \\ &= \frac{7}{10}\end{aligned}$$

#####

Running problem 2/15

#####

=====
=====

CONDITION B: Classification Only: Topic/domain metadata, no method card guidance

Problem ID: P02

Problem: Multiply 3/4 by 2/5

=====

Classification Signal from Elasticsearch

Score: 0.70102024 | Topic: e_equivalent_fractions | Domain: fractions-decimals | Doc: fractions-decimals__e_equivalent_fractions-p002

Score: 0.67506814 | Topic: m_mixed_number | Domain: fractions-decimals | Doc: fractions-decimals__m_mixed_number-p002

Score: 0.66949725 | Topic: f_fractions_multiplication | Domain: fractions-decimals | Doc: fractions-decimals__f_fractions_multiplication-p002

$$\begin{aligned}\frac{3}{4} \times \frac{2}{5} &= \frac{3 \times 2}{4 \times 5} \\ &= \frac{6}{20} \\ &= \frac{6 \div 2}{20 \div 2} \\ &= \frac{3}{10}\end{aligned}$$

#####

Running problem 3/15

#####

=====

CONDITION B: Classification Only: Topic/domain metadata, no method card guidance

Problem ID: P03

Problem: The fraction 3/4 written as a decimal is

=====

Classification Signal from Elasticsearch

Score: 0.7162106 | Topic: d_decimal_fraction_percentage | Domain: fractions-decimals |
Doc: fractions-decimals__d_decimal_fraction_percentage-p001

Score: 0.71258044 | Topic: c_converting_fractions_decimals | Domain: fractions-decimals |
Doc: fractions-decimals__c_converting_fractions_decimals-p001

Score: 0.7060766 | Topic: d_decimal | Domain: fractions-decimals | Doc: fractions-
decimals__d_decimal-p001

$$\frac{3}{4} = 3 \div 4 = 0.75$$

#####

#####

Running problem 4/15

#####

#####

=====

CONDITION B: Classification Only: Topic/domain metadata, no method card guidance

Problem ID: P04

Problem: If $R = \{a, b, c, d, e, f\}$, draw the set that is not a subset of R

=====

Classification Signal from Elasticsearch

Score: 0.67028904 | Topic: s_subset | Domain: set-theory | Doc: set-theory__s_subset-p002

Score: 0.668005 | Topic: i_intersection | Domain: set-theory | Doc: set-theory__i_intersection-p002

Score: 0.6619897 | Topic: p_proper_subset | Domain: set-theory | Doc: set-theory__p_proper_subset-p001

A set that is not a subset of R must contain at least one element that is not in R .
For example, let $S = \{a, b, g\}$.

Since 'g' is an element of S but not an element of R , S is not a subset of R .

#####

Running problem 5/15

#####

=====
=====

CONDITION B: Classification Only: Topic/domain metadata, no method card guidance

Problem ID: P05

Problem: Draw a rectangular shape that shows the shading of $\frac{1}{3}$

=====
=====

Classification Signal from Elasticsearch

Score: 0.6185677 | Topic: f_fraction_number_line | Domain: fractions-decimals | Doc:
fractions-decimals__f_fraction_number_line-p001

Score: 0.61370826 | Topic: d_decimal_fraction_percentage | Domain: fractions-decimals |
Doc: fractions-decimals__d_decimal_fraction_percentage-p003

Score: 0.6094401 | Topic: c_converting_fractions_decimals | Domain: fractions-decimals |
Doc: fractions-decimals__c_converting_fractions_decimals-p002

Fraction Bar



$\frac{1}{3}$ shaded

#####

Running problem 6/15

#####

=====
=====

CONDITION B: Classification Only: Topic/domain metadata, no method card guidance

Problem ID: P06

Problem: Show that $3 * 7 + 3 * 9$ is the same as $7 + 7 + 7 + 9 + 9 + 9$

=====
=====

Classification Signal from Elasticsearch

Score: 0.62641335 | Topic: e_exponentiation | Domain: arithmetic-operations | Doc: arithmetic-operations__e_exponentiation-p001

Score: 0.6248069 | Topic: p_product | Domain: arithmetic-operations | Doc: arithmetic-operations__p_product-p001

Score: 0.62218285 | Topic: s_sum | Domain: arithmetic-operations | Doc: arithmetic-operations__s_sum-p001

$$\begin{aligned} 3 \cdot 7 + 3 \cdot 9 &= 21 + 27 \\ &= 48 \end{aligned}$$

$$\begin{aligned} 7 + 7 + 7 + 9 + 9 + 9 &= 3 \cdot 7 + 3 \cdot 9 \\ &= 21 + 27 \\ &= 48 \end{aligned}$$

Since both expressions simplify to 48, they are the same.

#####

Running problem 7/15

#####

=====
=====

CONDITION B: Classification Only: Topic/domain metadata, no method card guidance

Problem ID: P07

Problem: In a proper fraction, the difference between the numerator and denominator is 5.
If the denominator is 7, what is the fraction?

=====
=====

Classification Signal from Elasticsearch

Score: 0.77135897 | Topic: p_proper_fraction | Domain: fractions-decimals | Doc:
fractions-decimals__p_proper_fraction-p002

Score: 0.76488376 | Topic: p_proper_fraction | Domain: fractions-decimals | Doc:
fractions-decimals__p_proper_fraction-p001

Score: 0.71748185 | Topic: p_proper_fraction | Domain: fractions-decimals | Doc:
fractions-decimals__p_proper_fraction-p003

Let the fraction be $\frac{n}{d}$.

Given that the denominator is 7, so $d = 7$.

The difference between the numerator and denominator is 5, so $d - n = 5$ (since it's a proper fraction, $d > n$).

$$7 - n = 5$$

$$n = 7 - 5$$

$$n = 2$$

Therefore, the fraction is $\frac{2}{7}$.

#####

Running problem 8/15

#####

=====
=====

CONDITION B: Classification Only: Topic/domain metadata, no method card guidance

Problem ID: P08

Problem: What is 1/8 of 16

=====

Classification Signal from Elasticsearch

Score: 0.59877014 | Topic: c_converting_fractions_decimals | Domain: fractions-decimals |
Doc: fractions-decimals__c_converting_fractions_decimals-p002

Score: 0.59734344 | Topic: t_thousandth | Domain: fractions-decimals | Doc: fractions-
decimals__t_thousandth-p001

Score: 0.5958979 | Topic: f_fraction_decimal_chart | Domain: fractions-decimals | Doc:
fractions-decimals__f_fraction_decimal_chart-p002

$$\begin{aligned}\frac{1}{8} \text{ of } 16 &= \frac{1}{8} \times 16 \\ &= \frac{1 \times 16}{8} \\ &= \frac{16}{8} \\ &= 2\end{aligned}$$

#####

#####

Running problem 9/15

#####

#####

=====

CONDITION B: Classification Only: Topic/domain metadata, no method card guidance

Problem ID: P09

Problem: The sum of 0.5 and 1.7 is

=====

Classification Signal from Elasticsearch

Score: 0.6196549 | Topic: m_mixed_number | Domain: fractions-decimals | Doc: fractions-decimals__m_mixed_number-p002

Score: 0.61845994 | Topic: s_sum | Domain: arithmetic-operations | Doc: arithmetic-operations__s_sum-p001

Score: 0.6165011 | Topic: i_improper_fraction | Domain: fractions-decimals | Doc: fractions-decimals__i_improper_fraction-p001

$$0.5 + 1.7 = 2.2$$

#####

#####

Running problem 10/15

#####

#####

=====

CONDITION B: Classification Only: Topic/domain metadata, no method card guidance

Problem ID: P10

Problem: The top length is a square is 7cm, what is the perimeter?

=====

Classification Signal from Elasticsearch

Score: 0.6843953 | Topic: p_perimeter | Domain: area-perimeter | Doc: area-perimeter__p_perimeter-p002

Score: 0.66611886 | Topic: m_mensuration | Domain: area-perimeter | Doc: area-perimeter__m_mensuration-p001

Score: 0.6581342 | Topic: p_perimeter | Domain: area-perimeter | Doc: area-perimeter__p_perimeter-p001

A square has four equal sides.

Given side length = 7 cm

Perimeter = $4 \times$ side length

= 4×7 cm

= 28 cm

#####

#####

Running problem 11/15

#####

=====
=====

CONDITION B: Classification Only: Topic/domain metadata, no method card guidance

Problem ID: P11

Problem: Show the largest to the smallest fraction and why they are so, $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{3}$, $\frac{1}{5}$

=====
=====

Classification Signal from Elasticsearch

Score: 0.70217276 | Topic: e_egyptian_fraction | Domain: fractions-decimals | Doc: fractions-decimals__e_egyptian_fraction-p001

Score: 0.69676995 | Topic: u_unit_fraction | Domain: fractions-decimals | Doc: fractions-decimals__u_unit_fraction-p002

Score: 0.6870134 | Topic: s_simplifying_fractions | Domain: fractions-decimals | Doc: fractions-decimals__s_simplifying_fractions-p002

To compare fractions, we can find a common denominator.

The given fractions are $\frac{1}{2}, \frac{1}{4}, \frac{1}{3}, \frac{1}{5}$.

The least common multiple (LCM) of the denominators 2, 4, 3, and 5 is 60.

Convert each fraction to an equivalent fraction with a denominator of 60:

$$\frac{1}{2} = \frac{1 \times 30}{2 \times 30} = \frac{30}{60}$$

$$\frac{1}{4} = \frac{1 \times 15}{4 \times 15} = \frac{15}{60}$$

$$\frac{1}{3} = \frac{1 \times 20}{3 \times 20} = \frac{20}{60}$$

$$\frac{1}{5} = \frac{1 \times 12}{5 \times 12} = \frac{12}{60}$$

Now, compare the numerators of the equivalent fractions:

$$30 > 20 > 15 > 12$$

Therefore, the fractions from largest to smallest are:

$$\frac{30}{60} > \frac{20}{60} > \frac{15}{60} > \frac{12}{60}$$

Which means:

$$\frac{1}{2} > \frac{1}{3} > \frac{1}{4} > \frac{1}{5}$$

#####

Running problem 12/15

#####

=====

CONDITION B: Classification Only: Topic/domain metadata, no method card guidance

Problem ID: P12

Problem: Draw $A \cup B$ if $A = \{1, 5, 6\}$, $B = \{a, c, d\}$

=====

Classification Signal from Elasticsearch

Score: 0.6957648 | Topic: c_cup | Domain: set-theory | Doc: set-theory__c_cup-p002

Score: 0.6882458 | Topic: u_union | Domain: set-theory | Doc: set-theory__u_union-p001

Score: 0.6838386 | Topic: u_union | Domain: set-theory | Doc: set-theory__u_union-p002

Given: $A = \{1, 5, 6\}$ $B = \{a, c, d\}$
 $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$
 $= \{1, 5, 6, a, c, d\}$

#####

Running problem 13/15

#####

=====

CONDITION B: Classification Only: Topic/domain metadata, no method card guidance

Problem ID: P13

Problem: Set A = {Even prime numbers greater than 2}, how many members does set A have?

=====

Classification Signal from Elasticsearch

Score: 0.6831758 | Topic: s_set_braces | Domain: set-theory | Doc: set-theory__s_set_braces-p001

Score: 0.67578673 | Topic: c_countable | Domain: set-theory | Doc: set-theory__c_countable-p001

Score: 0.6723573 | Topic: e_element_of_a_set | Domain: set-theory | Doc: set-theory__e_element_of_a_set-p001

A prime number is a natural number greater than 1 that has no positive divisors other than 1 and itself.

The first few prime numbers are 2, 3, 5, 7, 11, ...

An even number is an integer that is divisible by 2.

The only even prime number is 2.

The problem asks for even prime numbers greater than 2.

Since 2 is the only even prime number, there are no even prime numbers greater than 2.

Therefore, Set A is an empty set, denoted as $\emptyset$ or $\{\}$.

The number of members in Set A is 0.

#####

#####

Running problem 14/15

#####

=====
=====

CONDITION B: Classification Only: Topic/domain metadata, no method card guidance

Problem ID: P14

Problem: Draw a rectangle having 6 regions inside, with 1/2 shaded

=====
=====

Classification Signal from Elasticsearch

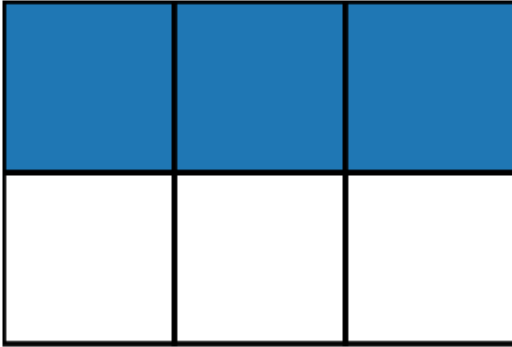
Score: 0.6007097 | Topic: q_quotient | Domain: arithmetic-operations | Doc: arithmetic-operations__q_quotient-p003

Score: 0.6007097 | Topic: s_sum | Domain: arithmetic-operations | Doc: arithmetic-operations__s_sum-p003

Score: 0.60047984 | Topic: a_area_irregular_polygons | Domain: area-perimeter | Doc: area-perimeter__a_area_irregular_polygons-p001

$$\frac{1}{2} \text{ of } 6 = 3$$

Rectangle with $\frac{1}{2}$ shaded



3 out of 6 regions shaded

#####

Running problem 15/15

#####

=====
=====

CONDITION B: Classification Only: Topic/domain metadata, no method card guidance

Problem ID: P15

Problem: The mixed number $7\frac{2}{5}$ written as an improper fraction is

=====
=====

Classification Signal from Elasticsearch

Score: 0.7374382 | Topic: m_mixed_number | Domain: fractions-decimals | Doc: fractions-decimals__m_mixed_number-p003

Score: 0.7227895 | Topic: m_mixed_number | Domain: fractions-decimals | Doc: fractions-decimals__m_mixed_number-p001

Score: 0.72033834 | Topic: i_improper_fraction | Domain: fractions-decimals | Doc: fractions-decimals__i_improper_fraction-p001

$$\begin{aligned} 7\frac{2}{5} &= \frac{7 \times 5 + 2}{5} \\ &= \frac{35 + 2}{5} \\ &= \frac{37}{5} \end{aligned}$$

#####

FINISHED CONDITION B TEST RUN

#####